

Nano-confined ionic liquid [emim][PF₆] between graphite sheets: A molecular dynamics study

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ABSTRACT

We have presented molecular dynamics (MD) simulations on the IL [emim][PF₆] confined between the graphite sheets with the different distances of 1, 2, 3, 4, and 5 nm. The vertical and longitudinal pressures and compressibility factors have been calculated and two separate equations of state for the vertical and longitudinal directions for the confined system have been derived by considering the anisotropic pressure tensor.

We have also investigated the effects of pore size and pore loading on the structural and dynamical properties of the confined ions. Our structural results showed a significant layering at the IL-graphite interface which indicates strong interaction between the IL and the graphite sheets. Our radial distribution function (RDF) results showed that the cation–anion RDF is greater than the other RDFs, indicating a strong interaction between cations and anions which is responsible for the H-bond formation. The average number of the hydrogen bonds also increased as the pore size decreased. The average number of the H-bonds also increases as the pore loading increases. Our dynamical results showed that the self diffusion coefficient of both cations and anions increases with increase in pore size. Also, the self diffusion of the confined ions decreases with increasing the pore loading.

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1. Introduction

The study of ionic liquids (ILs) in a confined state presents an exciting challenge and is of importance in practical applications [1]. Such a hybrid material (an inorganic porous material filled with a IL) represents a novel class of materials with promising applications [2]. Owing to the high specific surface area of the porous material and the peculiar properties of IL (such as low volatility, low flammability, high ionic conductivity, and high thermal stability) [3–8] these materials have important applications such as energy-storage devices (electrochemical batteries, supercapacitors) [9–11], lubricants [12], and drug delivery systems [2,4,13,14]. The understanding of the phase behavior of confined ILs has also attracted great attention due to their relevance in catalytic processes, lubricants, nanocomposites, fuels, and solar cells [15–22]. Many interesting phenomena have been also reported regarding the effects of nanoconfinement on changes in the melting point, crystal structure, and chemical reactivity of confined species [1,2,16,18,21,23–28]. For example, Kanakubo et al. [16] reported an intriguing melting-point-depression of 1,3-dialkylimidazolium-based ionic liquids confined in controlled pore glasses. Chen et al. [18] also found a transition of [bmim][PF₆] from liquid to a high melting-point crystal when it was confined in multiwalled CNTs. Sha et al. [21] reported a drastic phase transition in [dmim][Cl] ionic liquid confined between two graphene walls. Singh et al. [27] simulated the structural and

dynamical properties of the ionic liquid (IL) [bmim][PF₆] confined inside multiwalled carbon nanotubes (MWCNTs). They found that the diameter of the MWCNT and the pore loading have a profound influence on the structural and dynamical properties of the confined IL. Feng et al. [28] carried out molecular dynamics (MD) simulations to investigate the solid–liquid transition of ionic liquid 1-ethyl-3-methyl imidazolium bromide ([emim][Br]) by different approaches. They achieved a coexisting solid–liquid system in the microcanonical ensemble method. Recently, Li et al. [26] confined [emim][Br] ionic liquid in porous SiO₂ nanoparticles. They showed that the confinement in the nanopores leads to a compression of molecular size and an increase in the melting point. Also, Ori et al. [2] performed MD simulations in the isobaric–isothermal ensemble to investigate the structure and dynamics of an ionic liquid confined at ambient temperature and pressure in hydroxylated amorphous silica nanopores. They found that the local structure of the confined ionic liquid is mostly driven by electrostatic interactions. More recently, Akbarzadeh et al. [1] performed MD simulations to study the thermodynamics, structure, and dynamical behavior of 1-ethyl-3-methylimidazolium hexafluorophosphate [emim][PF₆] during the melting process inside carbon nanotubes (CNTs) with different radii.

However, little is understood about the morphology and phase behavior of the ILs in nanometer-scale confinement [22]. The actual structure of confined ILs is rarely understood at the atomic level and further experimental and theoretical investigations are expected to provide deep insight into the factors dominating the phase behavior of confined liquids [26].

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Table 1

The parameters of the fit to the extended EoS (Eqs. 5–7).

I	Y	Y_1	Y_2	Y_3
L	f	1.2434	0.2604	0.7703
	g	−3.9123	−0.4439	0.7568
	h	−0.0217	−0.0059	0.8047
H	f	0.8341	0.2939	0.5208
	g	−2.3755	−0.8759	0.3862
	h	0.0689	−0.0749	0.1889

The first goal of this paper is to determine two separate equations of state for the vertical and longitudinal directions for the IL confined between the graphite walls by considering the anisotropic pressure tensor. Such a fundamental investigation has not been performed before.

The second goal of this research is to understand how structural and dynamical properties of an IL [emim][PF₆] confined between the graphite walls are affected by the distance between the walls (the pore size effect). Such fundamental understanding is crucial to optimize the performance of several alternative energy devices, such as electrochemical double layer capacitors (EDLCs) and

dye-sensitized solar cells (DSSCs), as the molecular-level properties of confined ILs, affect the macroscopic performance of these devices [27].

2. Simulation details

The MD simulations have been performed on the IL [emim][PF₆] confined between the two graphite sheets with the different distances of 1, 2, 3, 4, and 5 nm. For each distance, several simulations with a different number of the IL molecules have been performed. All simulations have been carried out in the NVT ensemble with the Nosé–Hoover algorithm with the DL_POLY program version 4.03 [29] at 300 K. The equations of motion were integrated using the Verlet leapfrog algorithm and a time step of 1 fs. The relaxation time used for thermostat is 0.1 ps. Electrostatic interactions are calculated with the Ewald summation technique. The cutoff for long-range potentials is chosen as 15 Å. Periodic boundary conditions were applied in all three directions. To avoid artificial influence from periodic images, IL-graphite sheets were kept in the center of a simple orthorhombic box with vacuum on both sides separating it from the next periodic image in Z direction. Simulations have been run long enough to overcome the initial configurations. The simulations were carried out for 5 ns of equilibration

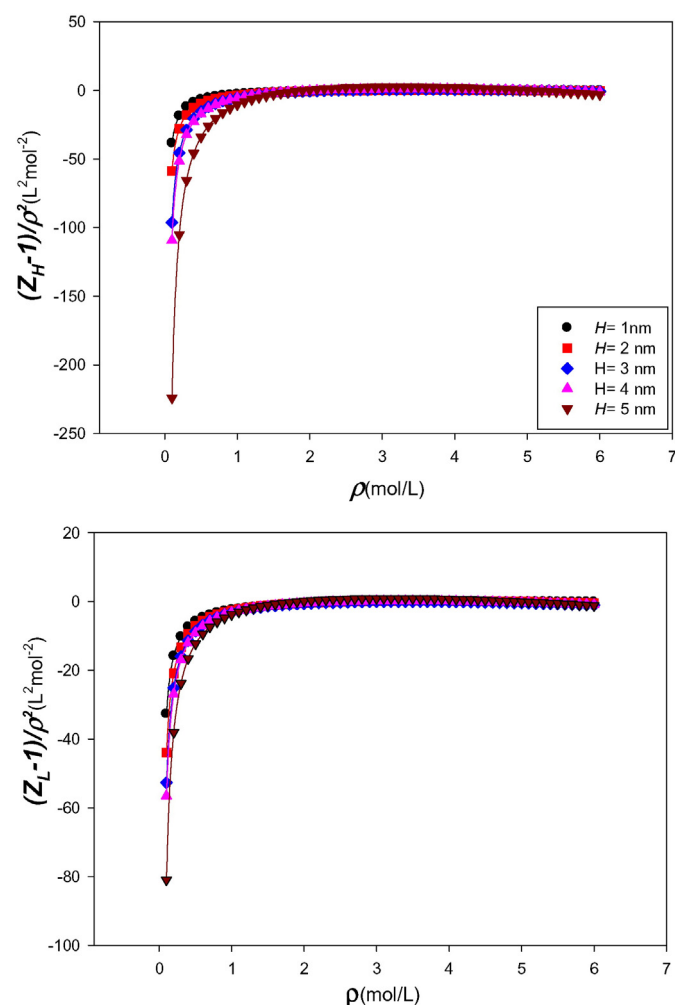


Fig. 1. The graphs of $(Z_H - 1) / \rho^2$ and $(Z_L - 1) / \rho^2$ versus density for the confined IL at different pore sizes which have been obtained from fitting to the simulation results.

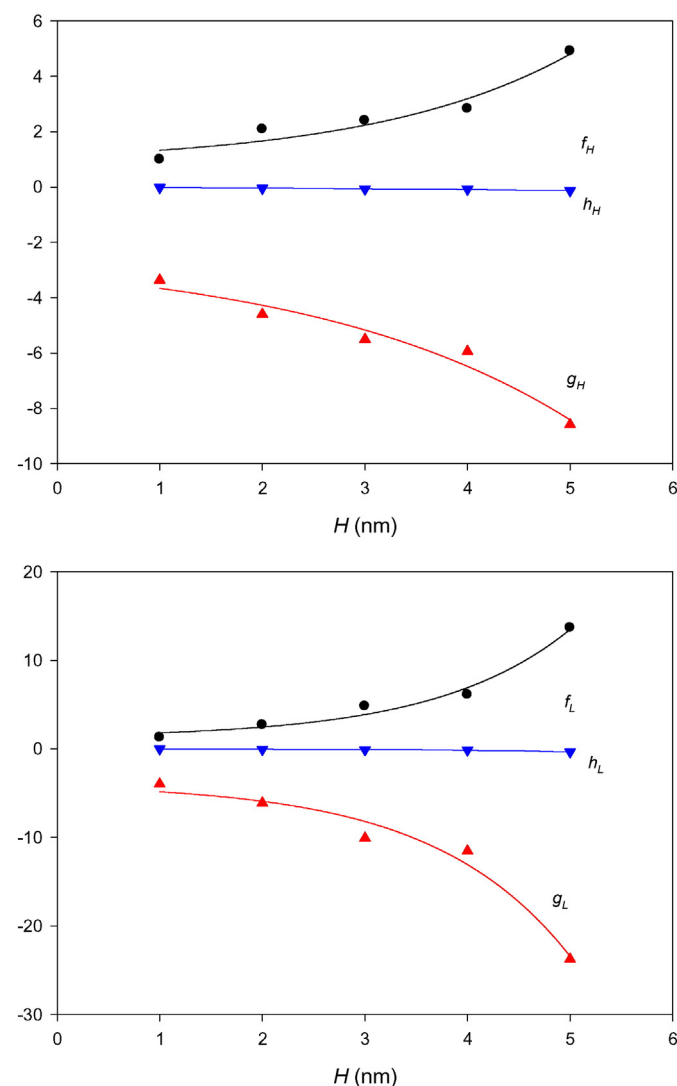


Fig. 2. The pore size dependence of the EoS parameters.

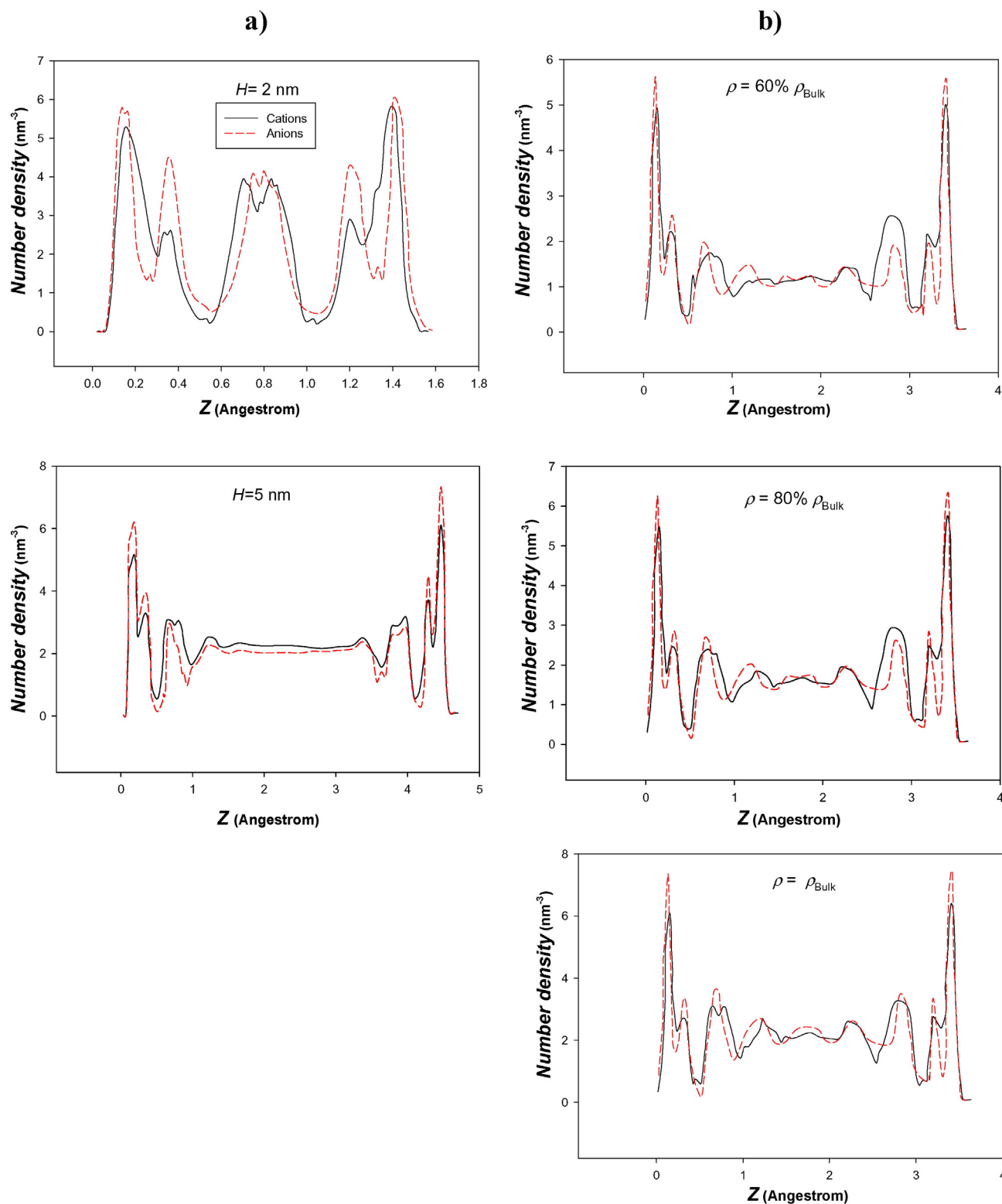


Fig. 3. a) The density profiles of the confined cations and anions at different pore sizes when the confined IL has a density similar to that of a bulk IL. b) The density profiles of the confined ions at different pore loadings (when the confined IL has 60%, 80%, and 100% of the bulk density at a fixed pore size ($H = 4$ nm)).

followed by production time of 40 ns for calculated properties. In our simulations, the slit pore was filled with a varying number of ion pairs so that our systems had a density similar to the bulk density at the temperatures of interest [30]. We also performed additional simulations at different degrees of pore loadings, where we reduced the number of ion pairs inside the graphite sheets.

In these simulations, the graphite sheets have been kept in fixed position. The force field parameters for the cations [emim] and anions [PF₆] were taken from Reference [31] as developed by Y. Zhang and E. J. Maginn which were successfully used to simulate the structures and dynamics of ionic liquids. Also, for the intermolecular interactions between ionic liquid and graphite sheet, we have used

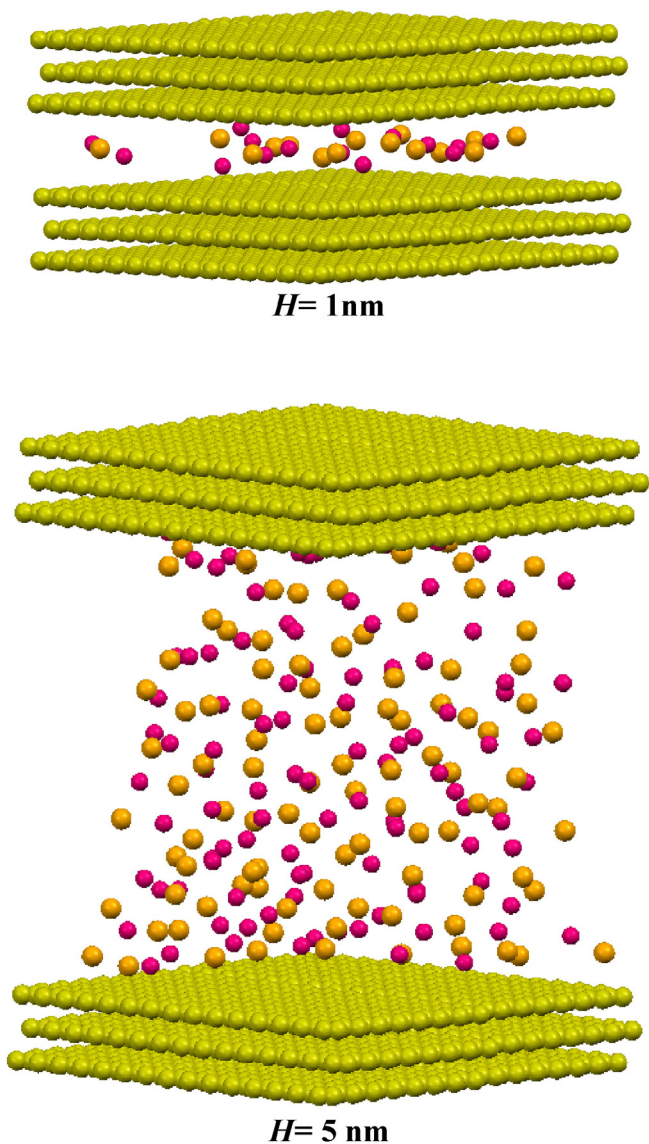


Fig. 4. The snapshots of the confined ionic liquids between the graphite sheets with the distances of 1 and 5 nm when the confined IL has a density similar to that of a bulk IL. Purple and yellow balls represent center-of-mass locations of [emim] and [PF₆], respectively. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Lennard–Jones potential utilizing the geometric mean for ε and the arithmetic mean for α .

3. Results and discussion

3.1. Equation of state (EoS)

The IL between the graphite wall is an anisotropic system for which the pressure tensor has to be used instead of a scalar pressure. The pressure along the graphite sheets (P_L) and in the vertical direction to the graphite sheets (P_H) directions can be obtained from the following relations [27]:

$$P_H = T \left(\frac{\partial P_H}{\partial T} \right)_{N,L,H} - \left(\frac{\partial U}{\partial V} \right)_{N,T,L} \quad (1)$$

$$P_L = T \left(\frac{\partial P_L}{\partial T} \right)_{N,L,H} - \left(\frac{\partial U}{\partial V} \right)_{N,T,H} \quad (2)$$

where U is the configurational energy and V is the molar volume. Following the notation used by Parsafar et al. [27,28] and using Eqs. 1 and 2, the exact thermodynamic EoS for the vertical and longitudinal components of pressure tensor may be rewritten as:

$$(Z_H - 1)V^2 = -\frac{1}{\rho^3 k_B T} \left(\frac{\partial U}{\partial V} \right)_{N,T,L} + \frac{1}{\rho^2} \left[\frac{1}{\rho k_B} \left(\frac{\partial P_H}{\partial T} \right)_{N,L,H} - 1 \right] \quad (3)$$

$$(Z_L - 1)V^2 = -\frac{1}{\rho^3 k_B T} \left(\frac{\partial U}{\partial V} \right)_{N,T,H} + \frac{1}{\rho^2} \left[\frac{1}{\rho k_B} \left(\frac{\partial P_L}{\partial T} \right)_{N,L,H} - 1 \right] \quad (4)$$

where ρ is the density and $Z_H = P_H V / k_B T$ and $Z_L = P_L V / k_B T$ are the vertical and longitudinal compressibility factor, respectively. Substituting the configurational energy proposed by Parsafar et al. [28], on the basis of the effective near-neighbor pair interactions of an extended Lennard–Jones (12,6,3) type, the equations of state can be rewritten as:

$$\frac{(Z_H - 1)}{\rho^2} = f_H + \frac{g_H}{\rho} + h_H \rho^2 \quad (5)$$

$$\frac{(Z_L - 1)}{\rho^2} = f_L + \frac{g_L}{\rho} + h_L \rho^2 \quad (6)$$

where the parameters f_H , g_H , h_H , f_L , g_L , and h_L are functions of the pore size (the distance between the graphite sheets (H)) at constant temperature as:

$$Y_i = y_{i1} + y_{i2} e^{y_{i3} H} \quad (7)$$

where Y represents f , g , or h and the subscript i represents H or L . The values of the parameters of Eq. (7) have been presented in Table 1. The vertical and longitudinal pressures have been calculated using Eqs. 1 and 2. In order to study the applicability of the extended EoS (Eqs. 5 and 6) for the IL confined between the graphite sheets, the vertical and longitudinal compressibility factors have been calculated and the values of $(Z_H - 1) / \rho^2$ and $(Z_L - 1) / \rho^2$ for the systems at the different distances of the two graphite walls have been shown versus the molar density in Fig. 1. According to this figure, there is very good correlation between the calculated values and the fitted EoS. Therefore, the extended EOS (Eqs. 5 and 6) gives very accurate predictions for the confined ILs between the graphite walls. The pore size dependences of the parameters of the extended EoS have been also presented in Fig. 2.

3.2. Structural properties

We have presented the density profiles of the confined cations and anions at different pore sizes when the confined IL has a density similar to that of a bulk IL in Fig. 3. We have also shown the density profiles of the confined ions when the confined IL has 60%, 80%, and 100% of the bulk density at a fixed pore size ($H = 4$ nm) in Fig. 3. According to Fig. 3, the cations and anions show a similar trend of local maxima near the graphite walls with higher density of the ions at the interface as compared to the bulk IL. Formation of the layers close to the graphite walls and strong oscillations for both cations and anions can be observed. The layering indicates strong interaction between the IL and the graphite sheets. The layering behavior near the wall has been also reported in the previous studies [1,5,32]. It is also shown that the more clear oscillations exist in the smaller pores ($H = 2$ nm in Fig. 3), however a uniform density distribution can be observed in the center for large ($H = 5$ nm) pore size. According to Fig. 3, the first peak for both ions can be observed at the same position from the graphite wall. This can be due to the fact that both ions have stacked at the same distance from the graphite wall. It is also shown in Fig. 3 that the number of peaks increases with increasing the pore size. This can be due to observing the number of ions in the interfacial layer per unit surface

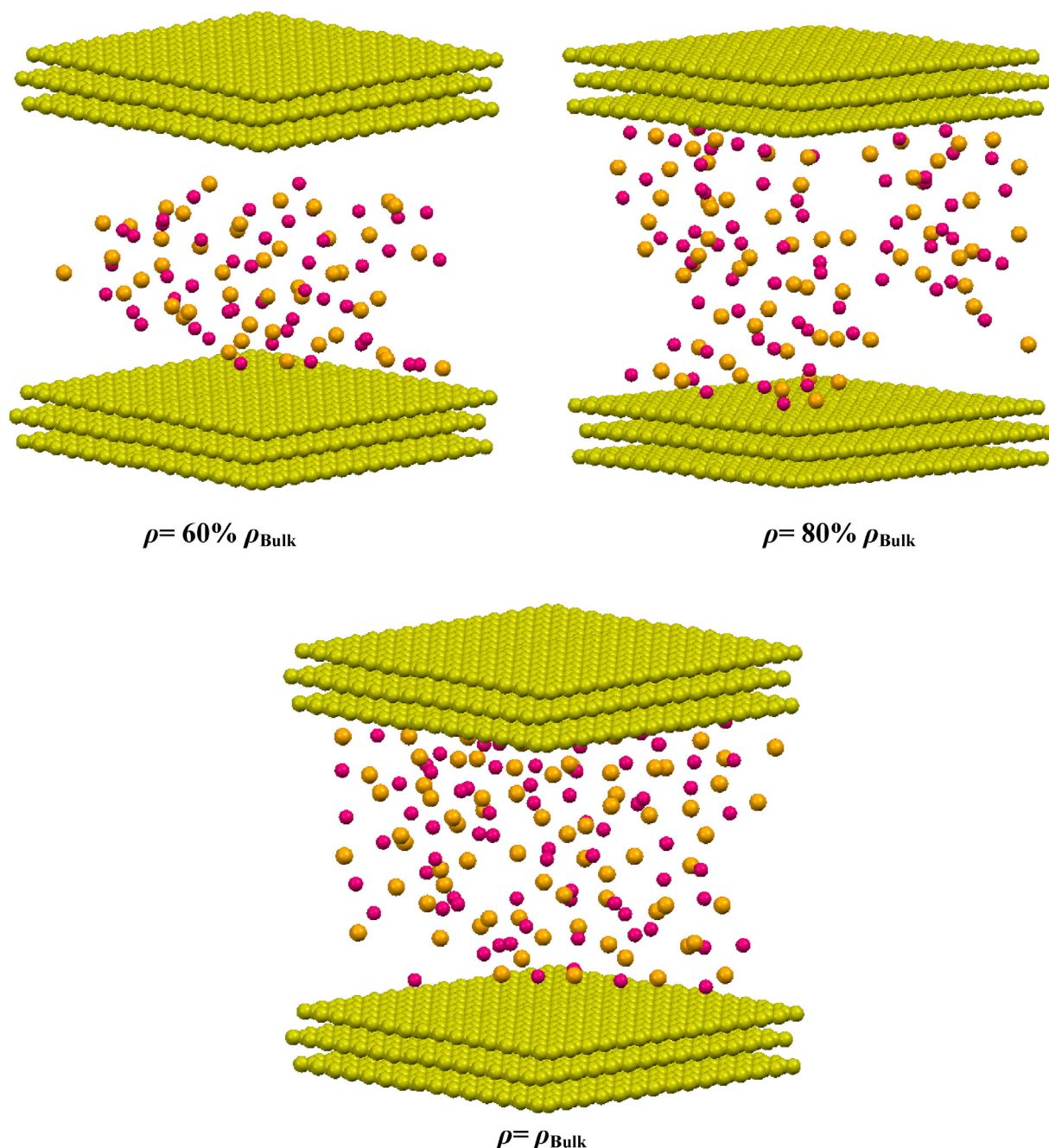


Fig. 5. The snapshots of the confined ILs between the graphite sheets when the confined IL has 60%, 80%, and 100% of the bulk density at fixed pore size.

area. The width of the peaks at the interface decreases with increasing the pore size, indicating decrease in the number of ions in the interfacial layer per unit surface area. The snapshots of the confined ionic liquids between the graphite sheets with the distances of 1 and 5 nm have been shown in Fig. 4 when the confined IL has a density similar to that of a bulk IL. We have also presented the snapshots of the confined ILs between the graphite sheets with the fixed distance of 4 nm in Fig. 5 when the confined IL has 60%, 80%, and 100% of the bulk density.

In order to investigate the orientations of the confined cations near the graphite walls, we have shown the average orientation of the confined cations between the graphite sheets with the distances of 2 and 5 nm in Fig. 6 when the confined IL has a density similar to that of a bulk IL. We have also presented the average orientation of the confined cations in Fig. 6 when the confined IL has 60%, 80%, and 100% of the bulk density at fixed pore size. In Fig. 6, the orientation of the cations has been measured by $\cos(\theta)$, where θ is the angle between the radial vector

and the vector normal to the imidazolium ring. An average value of about 0.5 for the $(\cos\theta)$ indicates no preferred orientation and a value close to one indicates that the imidazolium ring tend to align parallel to the surface which induced by the solid surface results in a dense region at the interface.

According to the first peaks in Fig. 6, the parameter $\cos(\theta)$ shows a very high value of about 0.8 which indicates that the adsorbed cations show high tendency to lie flat on the surface. It is also shown that some of the cations make a small angle with the surface. Therefore, cations can show multiple orientations at the surface. The result is in agreement with previous results [1,5,32]. Beneath this parallel layer of cations there is a low-density region with $\cos(\theta)$ of about 0.3 nm where cations are preferably perpendicular to the plane. Cations and anions are restrained in a plane configuration at the interface, which is due to the presence of the walls; this effect is somewhat enhanced by the strong interaction of graphite with cations and anions. No preferred

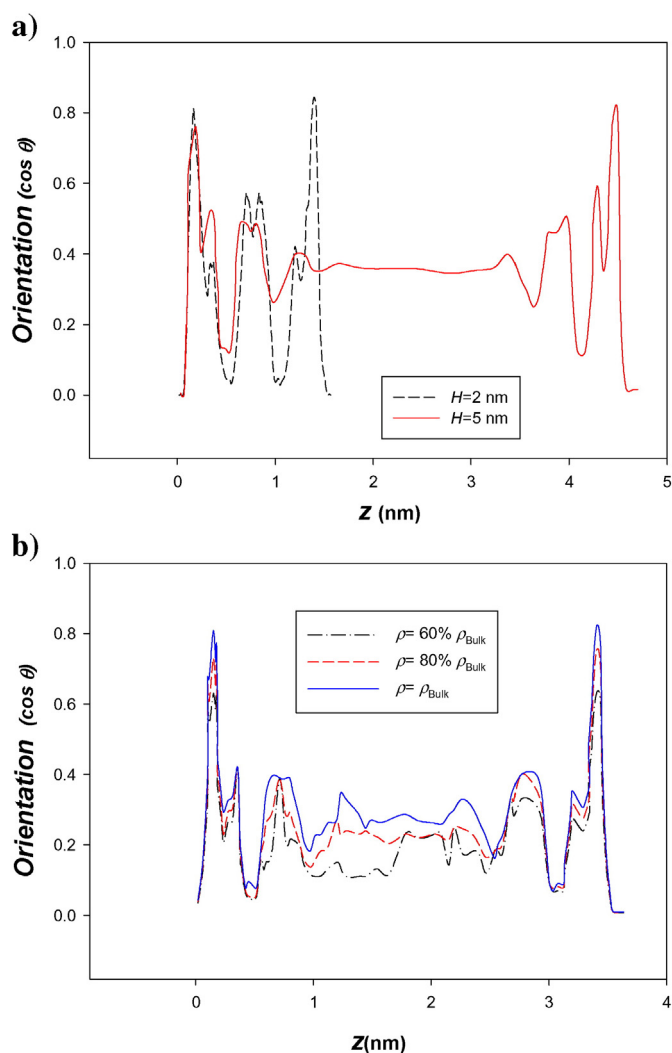


Fig. 6. a) The average orientation of the confined cations between the graphite sheets with the distances of 2 and 5 nm when the confined IL has a density similar to that of a bulk IL. b) The average orientation of the confined cations when the confined IL has 60%, 80%, and 100% of the bulk density at fixed pore size ($H = 4$ nm).

orientation order was observed as we move away from the surface or in the bulk because of loss of anisotropy. Confinement of ILs gives tighter packing and increased density of ions at the interface resulting in more favorable energy in confined systems as compared to the bulk liquid. According to Fig. 6, the corresponding plots for $H = 2$ and $H = 5$ nm are almost similar. It means that the ion orientation is independent of pore size.

To further investigate the local structure of the confined IL inside the slit pores, the radial distribution function (RDF) has been computed. We have presented the cation–graphite, anion–graphite, cation–cation, anion–anion, and cation–anion RDFs at two different pore sizes ($H = 1$ and 5 nm) in Fig. 7. According to this figure, the cation–anion RDF is greater than the other RDFs. The coordination shell for the distribution function between the cations and anions can be also observed at a shorter distance of 4.12 Å, indicating a strong interaction between cations and anions which is responsible for the H-bond formation. The effect of the pore size on the structure of the confined IL has been also shown in Fig. 7 by comparing the RDFs for the different pore sizes ($H = 1$ and 5 nm). The first peak of the cation–anion distribution has been observed at almost the same position for both pore sizes, indicating that the position of peaks is independent of pore size. However, with increasing the

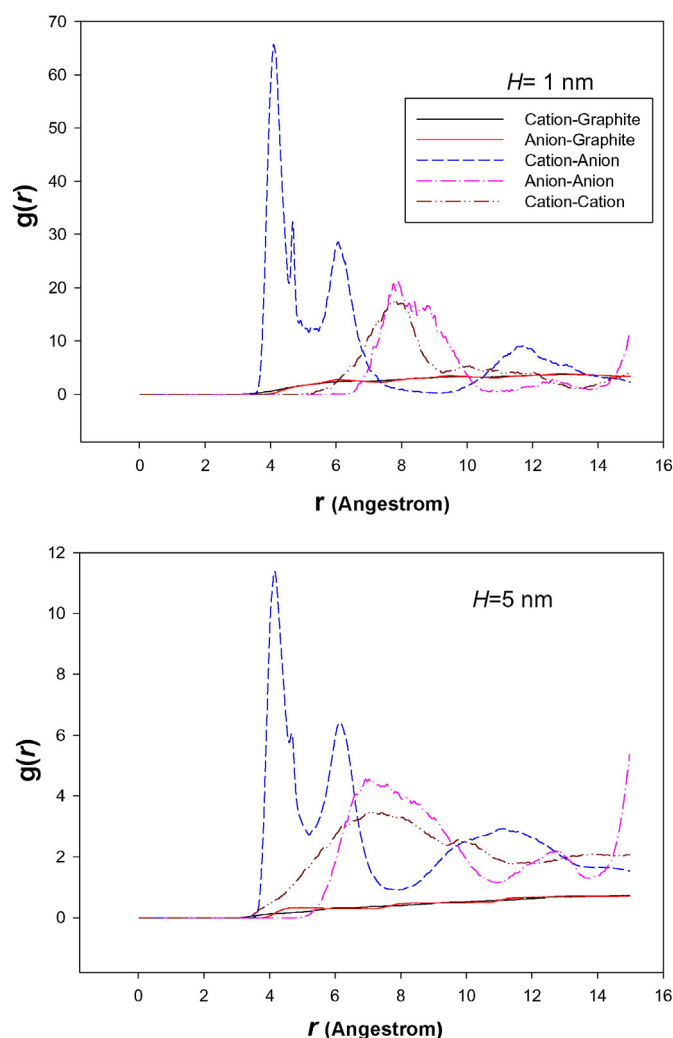


Fig. 7. The cation–graphite, anion–graphite, cation–cation, anion–anion, and cation–anion RDFs at two different pore sizes ($H = 1$ and 5 nm).

pore size, the height of the cation–anion RDF decreases. This effect may be due to confinement and excluded volume of systems [32–34]. The similar dependency on pore size has been observed in the other RDFs in Fig. 7.

We have also calculated the average number of the hydrogen bonds between the confined cations and anions at different pore sizes ($H = 1$ to 5 nm) and the different pore loadings (40% to 100% of the bulk density) in Fig. 8. (These results have been obtained using the calculation of the needed hydrogen bond length (2.67 Å) between the H atoms from the cations and the F atoms from the anions inside the nanotubes [35]). According to this figure, the average number of the hydrogen bonds increases as the pore size decreases. This result is in agreement with our RDF results in Fig. 7. In the other words, in the smaller pore size, the cations and anions have stronger interactions and so, the cation–anion RDF is greater and the more number of the H-bonds can be formed. It is also expected and shown in Fig. 8 that the average number of the hydrogen bonds increases as the pore loading increases.

3.3. Dynamical properties

The mobility of the confined ions in different directions is one of the factors in determining the internal resistance in EDLC, which ultimately affects its specific power [32,36]. By using the Einstein formula [37],

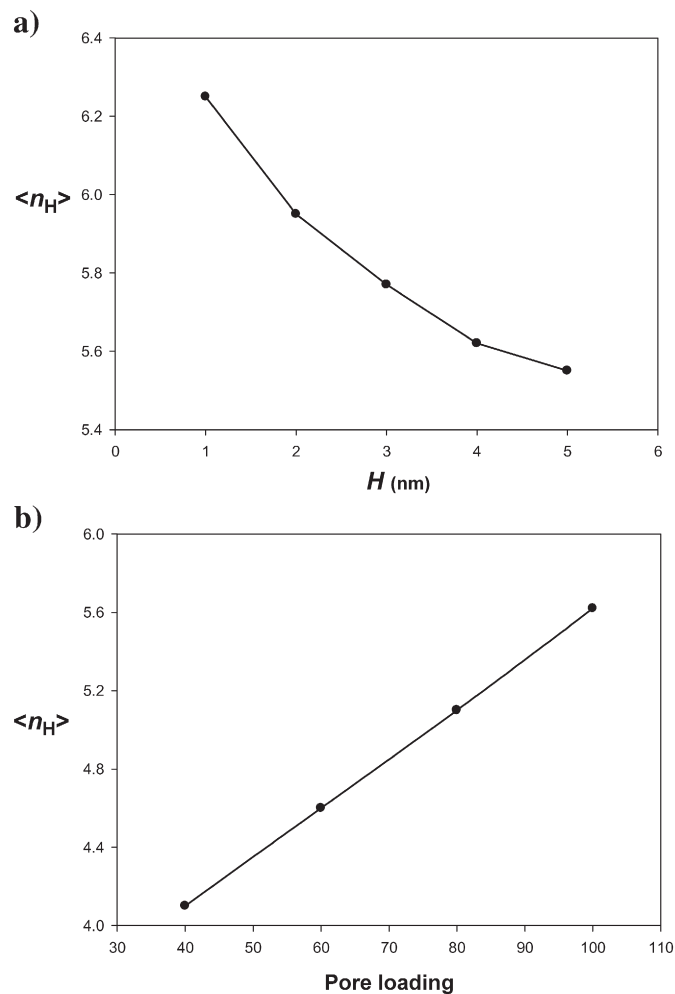


Fig. 8. a) The average number of the hydrogen bonds between the confined cations and anions at different pore sizes ($H = 1$ to 5 nm). b) The average number of the hydrogen bonds between the confined cations and anions at the different pore loadings (40% to 100% of the bulk density).

the self-diffusion coefficient, D , is obtained from the time-dependent mean-square displacement of particles, expected linear at large times:

$$D = \lim_{t \rightarrow \infty} \frac{1}{6tN} \sum_{i=1, N} \langle |r_i(t) - r_i(0)|^2 \rangle \quad (8)$$

where $r_i(0)$ and $r_i(t)$ are the positions of the particles at time $t = 0$ and t , respectively.

We have presented the self-diffusion coefficient for the confined cations and anions at different pore sizes ($H = 1$ to 5 nm) and the different pore loadings (40% to 100% of the bulk density) in Fig. 9.

According to Fig. 9, the self diffusion coefficient of both cations and anions increases with increase in pore size. This may be due to increase in fraction of ions which are away from the surface and have faster dynamics. It is also shown that the diffusion of the cations is faster than the anions. In spite of bulky and bigger size of [emim] than [PF₆], the cation shows larger value of diffusion coefficient. This observation can be due to the fact that the planar structure of imidazolium cations and the greater friction for translational motions of the anions as compared to the cations are responsible for faster diffusion of cations as compared to anions [32,38,39]. It is also shown in Fig. 9 that the self diffusion of the confined ions decreases with increasing the pore loading. This result is in agreement with the result of the average number of the hydrogen bonds in Fig. 9. It means that the average number of the

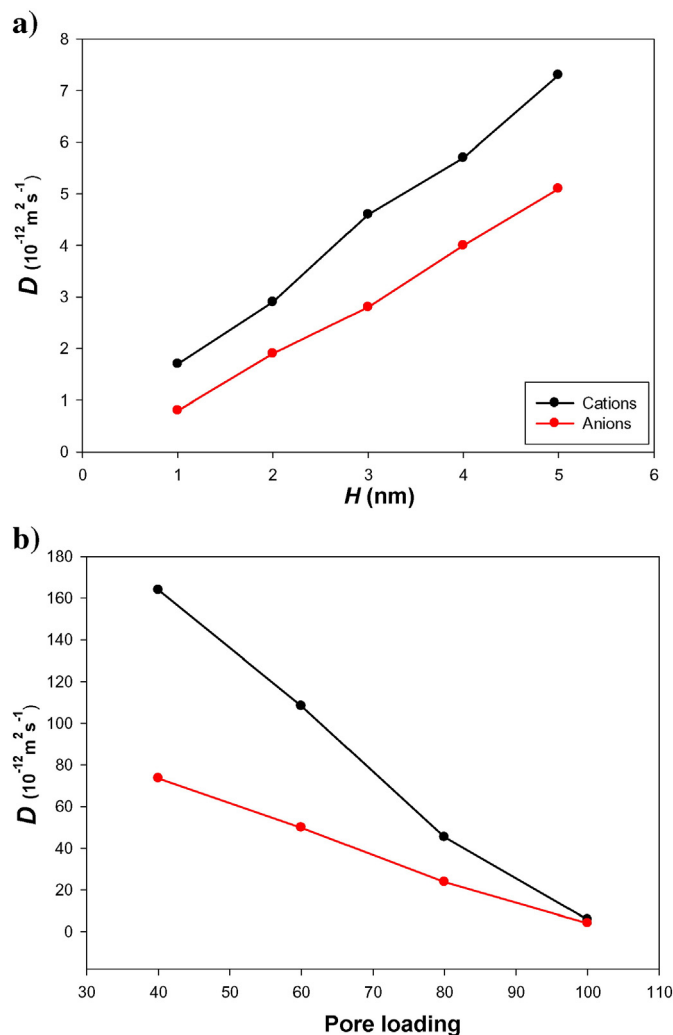


Fig. 9. a) The self-diffusion coefficient of the confined ions at different pore sizes ($H = 1$ to 5 nm). b) The self-diffusion coefficient of the confined ions at the different pore loadings (40% to 100% of the bulk density).

H-bonds increases as the pore loading increases and so, the dynamics of both cations and anions decreases.

4. Conclusions

The NVT-MD simulations have been performed on the IL [emim][PF₆] confined between the graphite sheets with the different distances of 1, 2, 3, 4, and 5 nm. For each distance, several simulations with a different number of the IL molecules have been performed. The vertical and longitudinal pressures and compressibility factors have been calculated and the recently proposed EoS of Sadeghi and Parasafar [27] in the vertical and longitudinal directions has been investigated and extended in terms of the distance between the graphite sheets.

The following structural and dynamical important results have been also summarized:

- 1- The density profiles for the confined ions show local maximums near the graphite walls with higher density of the ions at the interface. This indicates strong interaction between the IL and the graphite sheets.
- 2- The number of peaks in the density profiles increases with increasing the pore size. This can be due to observing the number of ions in the interfacial layer per unit surface area.

- 3- The cations can show multiple orientations at the surface. Beneath this parallel layer of cations there is a low-density region where cations are preferably perpendicular to the plane. It is also shown that the ion orientation is almost independent of pore size.
- 4- The cation–anion RDF is greater than the other RDFs, indicating a strong interaction between cations and anions which is responsible for the H-bond formation.
- 5- The height of the cation–anion RDF decreases with increasing the pore size. This effect is may be due to confinement and excluded volume of systems.
- 6- The average number of the hydrogen bonds increases as the pore size decreases. This result is in agreement with our RDF results. It is also expected and shown that the average number of the hydrogen bonds increases as the pore loading increases.
- 7- The self diffusion coefficient of both cations and anions increases with increase in pore size. This may be due to increase in fraction of ions which are away from the surface and have faster dynamics.
- 8- It is also shown that the diffusion of the cations is faster than the anions which is due to the planar structure of imidazolium cations and the greater friction for translational motions of the anions. Also, the self diffusion of the confined ions decreases with increasing the pore loading.

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